

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

SOME BASIC CONCEPT OF CHEMISTRY

LECTURE-4 CHEMISTRY

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Topics To Covered

1 ... Laws of Chemical Combination.

2 ... Daltons atomic Theory

3 ...

4 ...

REVISION



① Law of conservation of mass.



Total mass = Product mass.

2. Law of constant composition



$$12:32 = 3:8 \quad \checkmark$$

Source of Water



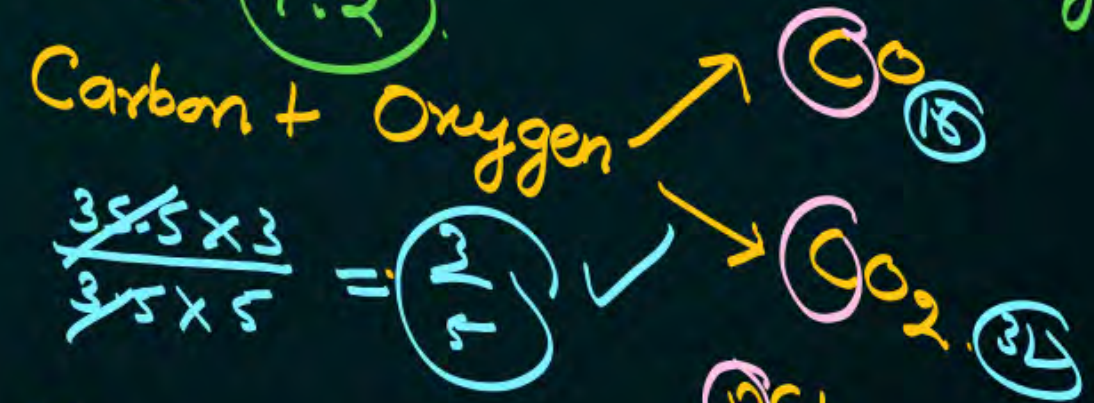
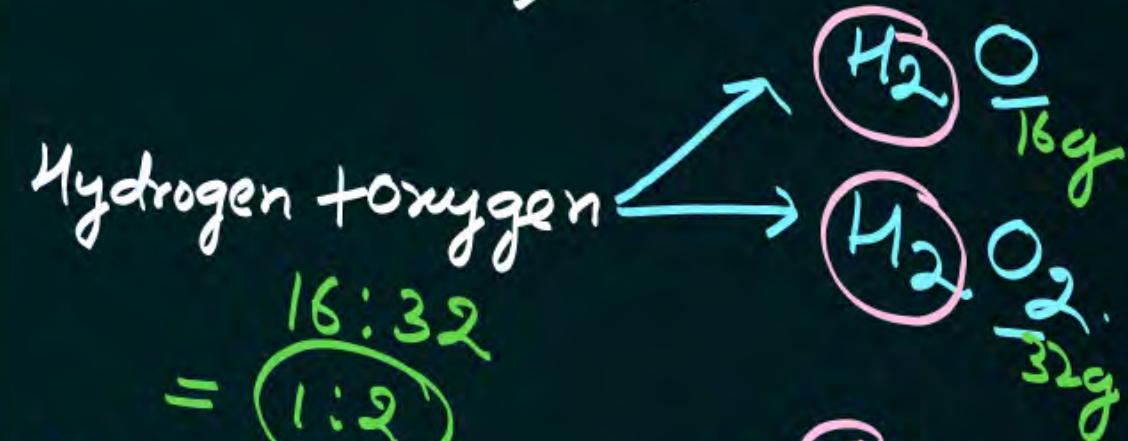
$$2:16 = 1:8$$

Number $\rightarrow$ 1, 2, 3, 4, 5, 6, 7, 8

$$(1.5) (2.4) (1.7) (2.9)$$

3) Law of multiple proportions
Same $\checkmark$.

When two different element



QUESTION



PVQ

^(CuO)
 1.375 g of cupric oxide was reduced by heating in a current of hydrogen and the weight of copper remained was 1.098 g. In another experiment, 1.179 g of copper was dissolved in HNO_3 and resulting copper nitrate converted into CuO by ignition the weight of CuO was 1.476 g. Show that these results, illustrate the law of constant composition.

Sol: $\rightarrow$ Case 1.

$$\text{Mass of CuO} = 1.375 \text{ g}$$

$$\text{Mass of Cu} = 1.098 \text{ g}$$

$$\begin{aligned} \text{Mass of Oxygen} &= 1.375 \text{ g} - 1.098 \text{ g} \\ &= 0.277 \text{ g} \end{aligned}$$

$$\therefore \text{Ratio of mass of Cu : O} = \frac{1.098}{0.277} = 3.96$$

Case 2.

$$\text{Mass of CuO} = 1.476 \text{ g}$$

$$\text{Mass of Cu} = 1.179 \text{ g}$$

$$\begin{aligned} \text{Mass of Oxygen} &= 1.476 - 1.179 \\ &= 0.297 \end{aligned}$$

$$\text{Ratio of mass of Cu : O} = \frac{1.179}{0.297} = 3.96$$

$$\begin{aligned} 1) \text{ CuO} &\rightarrow \frac{\text{Cu}}{\text{O}} = \text{same} \\ 2) \text{ CuO} &\rightarrow \frac{\text{Cu}}{\text{O}} = \text{same} \end{aligned}$$

Hence, following data, follows law of constant composition.

QUESTION

Carbon + Oxygen

(A)
(B)

Mass percentage

Carbon and oxygen are known to form two compounds. The carbon content in one of these compounds is 42.9% while in the other, it is 27.3%. Show that the data are in the agreement with the law of multiple proportions..

Sol: Case 1

Mass of Carbon = 42.9 g

Mass of Oxygen = $100 - 42.9$
 $= 57.1 \text{ g}$

42.9 g of Carbon combines with = 57.1 g of Oxygen

1 g of Carbon combines with = $\frac{57.1}{42.9}$ of Oxygen.
 $= 1.33 \text{ g}$

$$42.9 = 57.1$$

$$1 = \frac{57.1}{42.9}$$

Case 2

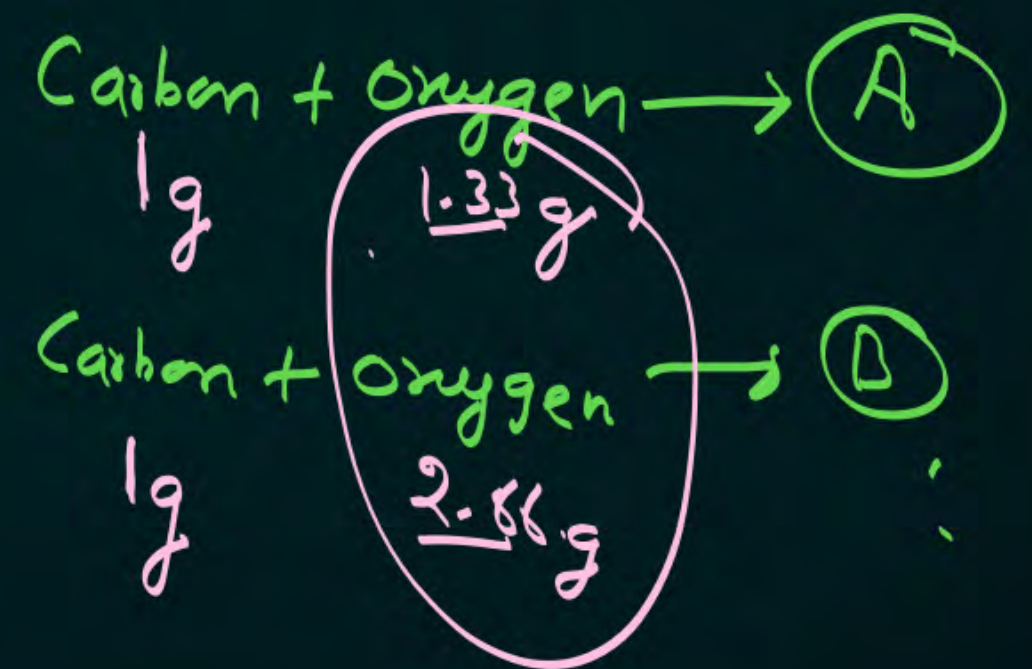
Mass of Carbon = 27.3 g

Mass of Oxygen = $100 - 27.3 = 72.7 \text{ g}$

27.3 g of Carbon combine with = 72.7 g of O

$$\begin{aligned} 1 \text{ g of Carbon} &= \frac{72.7}{27.3} \text{ of O} \\ &= 2.66 \text{ g} \end{aligned}$$

Therefore, Carbon has fixed mass of 1g & Oxygen has 1.33g & 2.66g in different cases. i.e. $\frac{1.33}{2.66} = \frac{1}{2}$ ratio or 1:2 ✓, hence law of multiple proportion is followed.



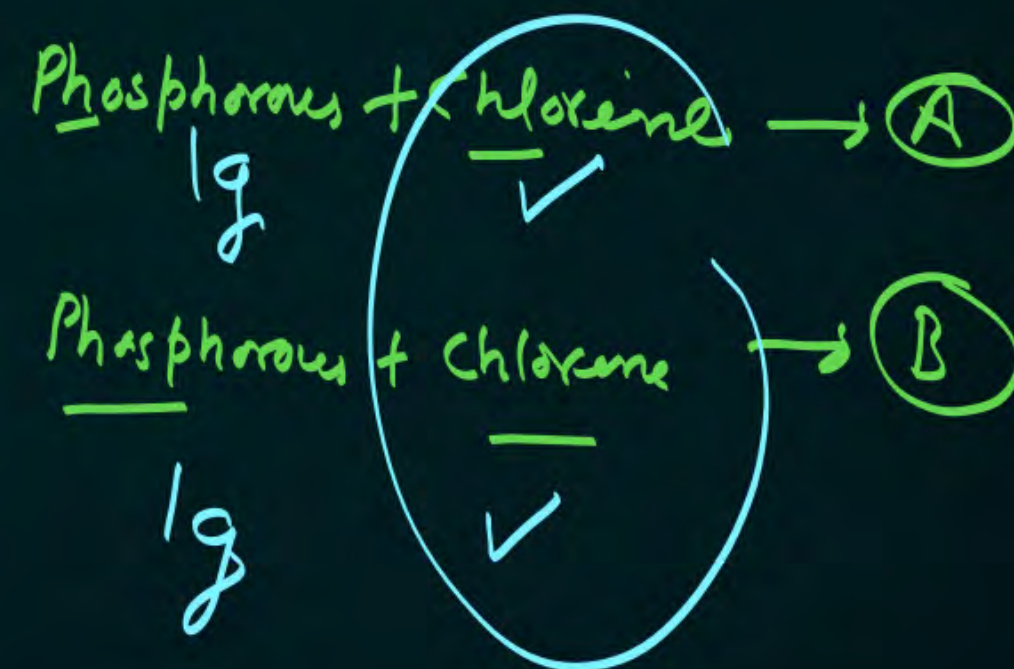
QUESTION



Home Work.

Phosphorus and chlorine form two compounds. The first compound contains 22.54% by mass of phosphorus and 77.46% by mass of chlorine. In the second compound the percentages are 14.88. for phosphorus and 85.12 for chlorine. Show that these data are consistent with the law of multiple proportions.

Hint: $\rightarrow$

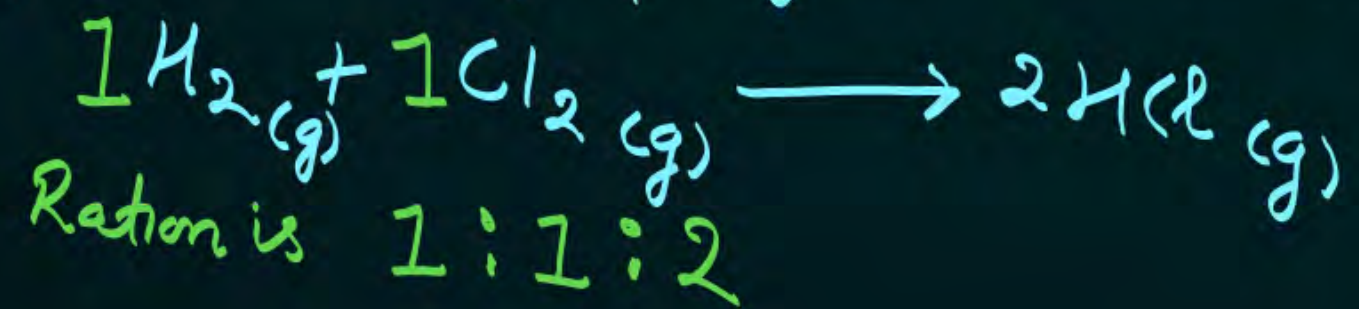


4) Gay Lussac's law of Combining Gases.

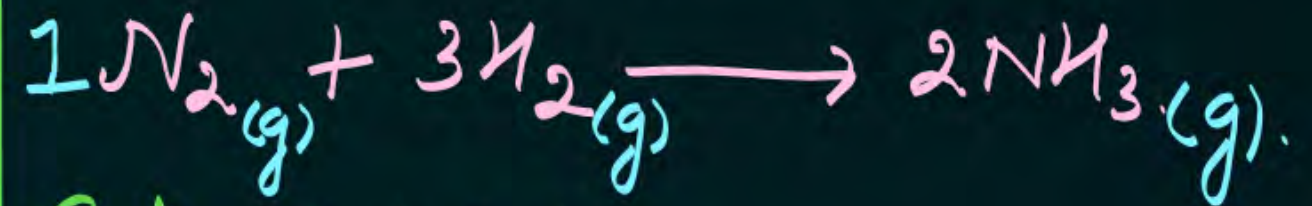
Given by Gay Lussac.

States that "under similar conditions of temp & pressure, whenever gases combine or produced in a chemical reaction, they do so in simple whole no. ratio by volume".

e.g a) Formation of HCl gas.



b) Formation of ammonia



Ratio 1 : 3 : 2

This is simple whole no. ratio.

5) Avagadro law $\Rightarrow$

Given by Amedeo Avagadro.

It states that "equal volume of gases, under similar conditions of temp and pressure, contain equal no. of molecules"

Mathematically, $V \propto N$

For example, if we take 1L of O_2 , 1L of N_2 & 1L of H_2 in three different containers.

Under similar condition of temp & pressure
Contain equal no. of molecules

O_2 - molecule ✓
 $O \rightarrow$ atom ✓



1L of O_2



1L of N_2



1L of H_2

Dalton's Atomic Theory →

To explain the

laws of ^{chemical combination} Dalton, proposed a theory. Assumptions are given below.

- 1) Matter is made up of extremely small indivisible, indestructible particle called atom.
- 2) Atoms of same elements have identical properties, i.e size, shape, ^{mass} chemical properties etc
- 3) Atoms of different elements have dissimilar properties. i.e shape, size, mass etc.
- 4) Atom is smallest particles, that involve the chemical reaction.
- 5) In chemical compounds different elements are always present in fixed ratio by mass.
- 6) Atom can neither be created nor be destroyed.
- 7) In a chemical reaction, there is combination, displacement & rearrangement of atoms.

QUESTION

2 : 1 : 2

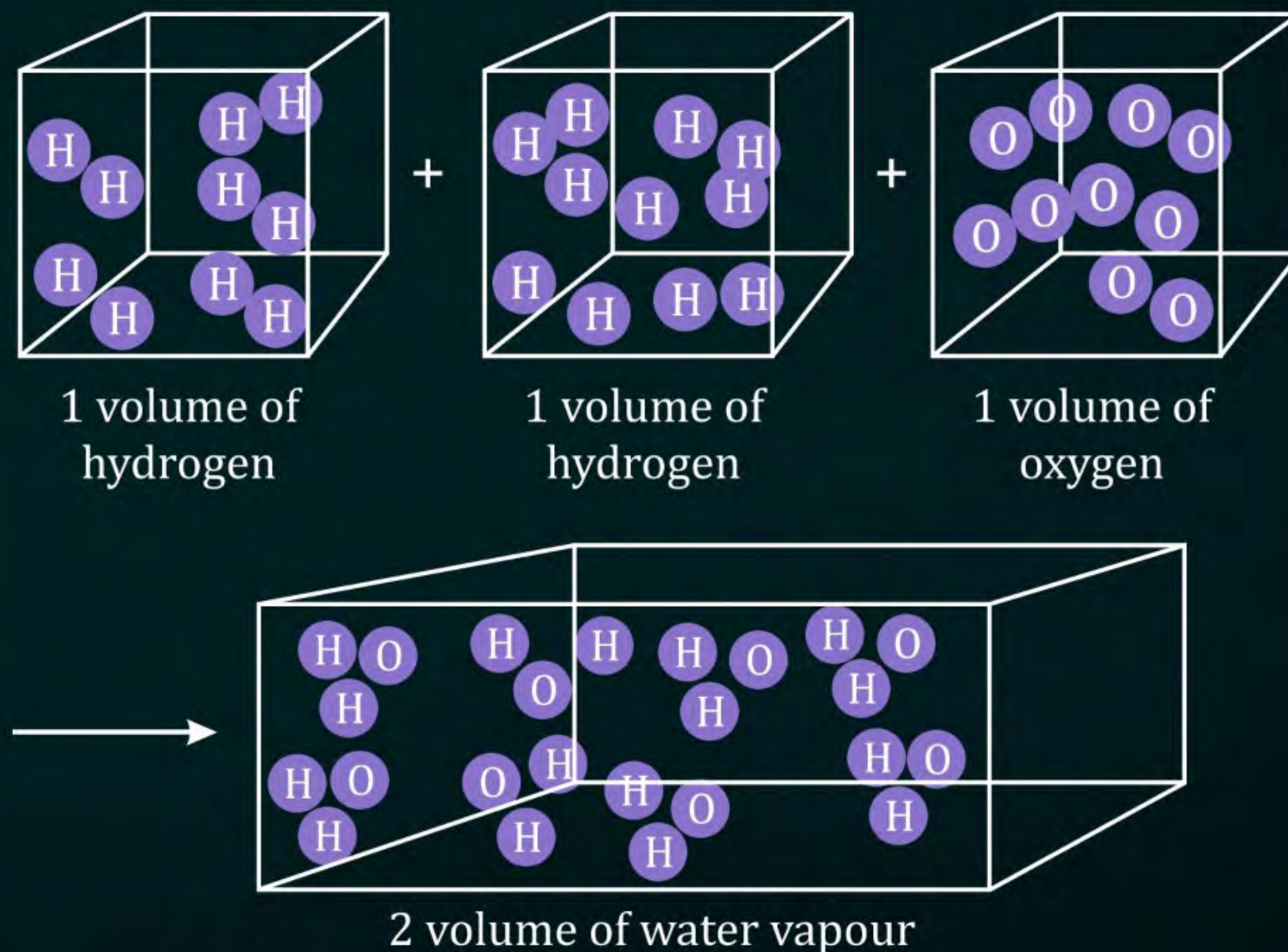
Which of the following laws of chemical combinations is satisfied by the figure?

1 Law of multiple proportion

2 Dalton's law

3 Gay Lussac's law of gaseous volume

4 Law of conservation of mass



REVISE KARNA

Thank
You